

k-Means Clustering

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Introduction

k -means is the workhorse of partitioning cluster analysis: assign each of n observations to one of k clusters, with the cluster centres as multivariate means and the objective being to minimise the total within-cluster sum of squared distances. It scales to large datasets, is conceptually simple, and produces interpretable centres in the original feature space. Its limitations — sensitivity to outliers, dependence on Euclidean geometry, the need to pre-specify k , and the assumption of approximately spherical equal-size clusters — are equally well known and motivate the alternatives (PAM, DBSCAN, Gaussian mixtures) in this category.

Prerequisites

A working understanding of distance metrics, multivariate means, and the trade-off between within-cluster compactness and between-cluster separation.

Theory

Lloyd's algorithm alternates two steps until convergence:

1. **Initialise** k centres, either randomly or via the k -means++ heuristic which spreads initial centres apart for better convergence.
2. **Assign** each observation to the cluster of its nearest centre.
3. **Update** each centre to the mean of its assigned observations.
4. **Repeat** until cluster assignments no longer change.

The objective is

$$\text{WSS} = \sum_{j=1}^k \sum_{x \in C_j} \|x - \bar{x}_j\|^2,$$

monotonically decreasing across iterations. The algorithm converges to a local optimum, so multiple random starts (`nstart >= 25`) are needed to find a good solution. Choice of k uses the elbow method, silhouette analysis, gap statistic, or composite tools like `NbClust`.

Assumptions

Continuous variables on comparable scales (standardise first); roughly spherical clusters of similar size; Euclidean distance is meaningful. Outliers can drag means and produce unbalanced clusters; PAM is preferable when outliers are present.

R Implementation

```
library(factoextra)

X <- scale(iris[, 1:4])
```

```

# 3 clusters
set.seed(2026)
km <- kmeans(X, centers = 3, nstart = 25)
km$size
km$centers

fviz_cluster(km, data = X, palette = "Set2", ggtheme = theme_minimal())

# Elbow and silhouette
fviz_nbclust(X, kmeans, method = "wss")
fviz_nbclust(X, kmeans, method = "silhouette")

```

Output & Results

`kmeans()` returns cluster assignments, cluster centres in the standardised feature space, total within-cluster sum of squares, between-cluster sum of squares, and convergence diagnostics. `fviz_cluster()` plots the clusters on the first two principal components with cluster ellipses; `fviz_nbclust()` automates elbow and silhouette diagnostics.

Interpretation

A reporting sentence: “ k -means with $k = 3$ and 25 random starts partitioned the standardised iris data with total within-cluster sum of squares 140; the resulting clusters closely matched the three species (overall accuracy 89 %, with versicolor and virginica contributing the misclassifications).” Always state k , the number of starts, and a quality metric (silhouette or WSS).

Practical Tips

- Always standardise predictors before k -means; variables with larger variance dominate Euclidean distance and bias the partition.
- Use `nstart = 25` or higher to avoid poor local optima; the algorithm is non-convex and a single random start can converge to a sub-optimal solution.
- For non-spherical clusters, switch to hierarchical clustering, DBSCAN (density-based), or Gaussian mixture models (model-based with elliptical clusters).
- For categorical data, k -means does not apply directly; use k -modes (`klaR::kmodes`) or PAM with Gower distance.
- Cross-validate the chosen k via stability measures (`fpc::clusterboot`); clusters that recur under bootstrapping are more trustworthy than those that change with each resample.
- Outliers can drag means; for outlier-contaminated data, prefer PAM (`cluster::pam`), trimmed k -means (`tclust::tkmeans`), or robust mixture models.

R Packages Used

Base R `kmeans()` for the standard algorithm; `factoextra` for `fviz_cluster()`, `fviz_nbclust()`, and ggplot-style diagnostics; `cluster` for related partitioning algorithms; `klaR::kmodes()` for categorical data; `fpc::clusterboot()` for stability-based cluster validation.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.

- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE